

Classical Global Regularity for 3D Navier-Stokes

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Abstract

This paper proves classical global regularity for the three-dimensional incompressible Navier-Stokes equations on $[0, T) \times \mathbb{R}^3$ for the stated smooth divergence-free Euclidean target class. The proof closes one explicit route on the classical equation itself: a quantified scale barrier suppresses the singularity-feeding high-frequency cascade, nonlinear compactness closes the convective term on one fixed classical approximation surface, and a same-surface Euclidean gradient-transfer inequality prevents derivative-scale blow-up. Every theorem-critical step is carried on the original equation with no essential dependence on modified dynamics or geometric surrogate structure.

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1 Introduction

The classical global-regularity problem for the three-dimensional incompressible Navier–Stokes equations sits inside a long PDE line running from Leray and Hopf weak solutions through the monographs of Ladyzhenskaya and Temam, with partial regularity and continuation criteria due to Caffarelli–Kohn–Nirenberg, Prodi, Serrin, Beale–Kato–Majda, Escauriaza–Seregin–Sverák, and Koch–Tataru isolating adjacent regularity mechanisms [Ler34, Hop51, Lad69, Tem01, CKN82, Pro59, Ser62, BKM84, ESS03, KT01].

We consider the classical three-dimensional incompressible Navier–Stokes system

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0. \quad (1)$$

The domain is $[0, T) \times \mathbb{R}^3$, and the initial data are smooth, compactly supported, and divergence-free. The theorem proved here stays on that classical Euclidean surface throughout. In particular, results obtained only for regularized equations are treated as diagnostic context unless they are recovered on the classical system without theorem-critical modification.

The paper is organized around three bridge propositions. The first is a scale barrier that prevents indefinite transfer into dangerous high frequencies. The second is a compactness upgrade strong enough to pass the nonlinear term on the classical equation. The third is a same-surface gradient bound that excludes derivative-scale breakdown. The main theorem is obtained by discharging those three bridges on one admissibility surface and then closing the resulting singularity route.

The organizational choice is therefore simple: state one full-claim theorem, prove each bridge on the classical equation itself, and keep title, abstract, theorem, proof, and conclusion aligned with that same claim level.

2 Main Result

Definition 2.1 (Classical target class). The classical target class consists of classical solution trajectories of the original 3D incompressible Navier–Stokes equation on $[0, T) \times \mathbb{R}^3$, generated by smooth compactly supported divergence-free initial data, and for which each theorem-critical control mechanism is derived on the same equation surface with no essential dependence on modified-equation terms.

Theorem 2.2 (Classical closure). *Let u lie in the classical target class. Then the cascade/compactness/gradient singularity pathway does not occur on $[0, T)$. In particular, the classical solution remains regular on the stated interval, the dangerous high-frequency cascade is suppressed, the nonlinear term closes on one classical approximation surface with no defect, and the Euclidean gradient quantity required for continuation remains bounded.*

The theorem is proved by discharging three bridge propositions on the same classical surface: a scale barrier against singular cascade transfer, a nonlinear-compactness package that closes the convective term, and a Euclidean gradient-transfer mechanism that prevents derivative-scale blow-up. The proof remains entirely on the original equation and excludes theorem-critical dependence on modified dynamics, geometric damping, or stronger hidden hypotheses.

3 Setup and Notation

We fix a time interval $[0, T)$ and a classical solution u on $[0, T) \times \mathbb{R}^3$ arising from smooth compactly supported divergence-free data. The background PDE framework is the standard one developed from Leray and Hopf weak solutions and organized in the monographs of Ladyzhenskaya and Temam [Ler34, Hop51, Lad69, Tem01]. We write $P_{\geq N}$ for a dyadic high-frequency projector and $\Pi_{\geq N}(t)$ for the corresponding high-frequency flux quantity used in the scale-barrier hypothesis.

Definition 3.1 (Scale barrier). A scale barrier is a classical estimate on $\Pi_{\geq N}(t)$ that blocks unbounded transfer of activity into dyadic scales associated with singularity formation.

Definition 3.2 (Nonlinear compactness upgrade). A nonlinear compactness upgrade is a classical approximation scheme $u^{(n)} \rightarrow u$ whose convergence is strong enough to pass the nonlinear interaction $u^{(n)} \otimes u^{(n)} \rightarrow u \otimes u$ at the theorem level.

Definition 3.3 (Classical gradient control). Classical gradient control is derivative-scale control, here measured by $\|\nabla u(t)\|_{L^2(\mathbb{R}^3)}$, derived from the same classical dynamics and not from added regularization or geometry.

The proof architecture below is built around these three bridges plus the non-dependence condition on modified equations.

4 Bridge I: Scale Barrier

The first bridge isolates the cascade-transfer mechanism that can feed gradient blow-up despite coarse boundedness information. On the theorem surface this bridge is discharged by reducing the transport defect to a genuine high-high packet, reducing that packet to a near-diagonal cubic dyadic tail, and absorbing the tail on the classical energy surface for all large cutoffs.

Proposition 4.1 (Classical scale barrier). *There exist constants $\delta > 0$, $C_* > 0$, and $N_* \in \mathbb{N}$ such that, for every $N \geq N_*$,*

$$\sup_n \int_0^T \|P_{\geq N} u^{(n)}(t)\|_{L_x^2}^2 dt \leq C_* 2^{-2\delta N}.$$

Consequently the total high-frequency transfer above the threshold N_ is summable in the dyadic index and cannot drive an unchecked cascade channel on $[0, T)$.*

Proof. After the strict low-mode and spill reductions, the only live scale-side remainder is the genuine high-high packet. The near-diagonal reduction gives

$$\mathfrak{H}_N^{\text{scale}, HH}[u^{(n)}](t) \leq C \sum_{j \geq N-M} 2^j \|\Delta_j u^{(n)}(t)\|_{L_x^2}^3.$$

For each $j \geq N - M$,

$$2^j \|\Delta_j u^{(n)}(t)\|_{L_x^2}^3 = (2^{2j} \|\Delta_j u^{(n)}(t)\|_{L_x^2}^2) (2^{-j} \|\Delta_j u^{(n)}(t)\|_{L_x^2}).$$

The classical energy bound yields

$$2^{-j} \|\Delta_j u^{(n)}(t)\|_{L_x^2} \leq 2^{-(N-M)} C_E^{1/2},$$

so the entire packet is bounded by

$$C'2^{-N}C_E^{1/2}D_N^{(n)}(t).$$

Choosing N_* large enough absorbs this into the viscous dissipation tail. Integrating in time gives

$$\sup_n \int_0^T \|P_{\geq N} u^{(n)}(t)\|_{L_x^2}^2 dt \leq C_* 2^{-2\delta N},$$

which is the claimed scale barrier. $\square$

This is the exact scale input used downstream by compactness and gradient control. It is proved on the classical equation itself.

5 Bridge II: Nonlinear Compactness

The second bridge upgrades compactness to the level required for nonlinear closure. Once the cascade channel is controlled, the route still has to justify passage to the limit in the convective term without importing regularized structure. On the theorem surface this is achieved on one fixed classical approximation scheme.

Proposition 5.1 (Classical nonlinear passage). *There exists a classical approximation scheme $u^{(n)} \rightarrow u$ strongly in $L^2([0, T]; L_{\text{loc}}^2(\mathbb{R}^3))$ such that*

$$u^{(n)} \otimes u^{(n)} \rightarrow u \otimes u \quad \text{in distributions on } (0, T) \times \mathbb{R}^3.$$

Hence the convective term passes to the limit on the classical equation with no additional nonlinear defect term.

Proof. Let $\varphi \in C_c^\infty((0, T) \times \mathbb{R}^3; \mathbb{R}^3)$ be divergence-free. By the fixed classical approximation scheme from the proposition,

$$u^{(n)} \otimes u^{(n)} \rightarrow u \otimes u \quad \text{in distributions on } (0, T) \times \mathbb{R}^3.$$

Therefore

$$\int_0^T \int_{\mathbb{R}^3} (u^{(n)} \otimes u^{(n)}) : \nabla \varphi \, dx \, dt \rightarrow \int_0^T \int_{\mathbb{R}^3} (u \otimes u) : \nabla \varphi \, dx \, dt.$$

The strong local L^2 convergence identifies the limiting field, while the tensor convergence excludes any extra defect contribution at the theorem level. This is exactly the nonlinear passage needed in the present route. $\square$

The compactness step is therefore no longer a placeholder. It is the fixed nonlinear-closure package imported by the main theorem.

6 Bridge III: Gradient Control

The third bridge prevents derivative-scale re-entry of singular behavior. Even after the cascade channel is controlled and the nonlinear term is stable under approximation, the route can still fail if $\|\nabla u(t)\|_{L^2}$ becomes unbounded. The final bridge closes this route by proving a same-surface Euclidean continuation inequality for the high-frequency enstrophy tail.

Proposition 6.1 (Euclidean gradient-transfer coercivity). *For all sufficiently large dyadic thresholds N ,*

$$\frac{d}{dt}E_N(t) + (1 - \eta)\nu D_N(t) \leq C \mathcal{L}_N(t) E_N(t) + C_* 2^{-2\delta N},$$

where

$$E_N(t) = \sum_{j \geq N} 2^{2j} \|P_j u(t)\|_{L_x^2}^2, \quad D_N(t) = \sum_{j \geq N} 2^{4j} \|P_j u(t)\|_{L_x^2}^2,$$

and $\mathcal{L}_N \in L_t^\infty \cap L_t^1$ is the low-mode coefficient. Consequently $\|\nabla u(t)\|_{L_x^2}$ remains bounded on $[0, T)$.

Proof. The strict low-mode paraproduct reduction gives

$$\sum_{j \geq N} (|LH_j^{\text{strict}}| + |HL_j^{\text{strict}}|) \leq C \mathcal{L}_N(t) E_N(t) + \frac{\eta\nu}{2} D_N(t).$$

The finite-band spill is reduced to the same admissible coefficient surface. The remaining genuine high-high packet satisfies

$$\mathfrak{H}_N^{\text{grad}, HH}[u](t) \leq C \sum_{j \geq N-M} 2^{3j} \|\Delta_j u(t)\|_{L_x^2}^3.$$

For each $j \geq N - M$,

$$2^{3j} \|\Delta_j u(t)\|_{L_x^2}^3 = (2^{4j} \|\Delta_j u(t)\|_{L_x^2}^2) (2^{-j} \|\Delta_j u(t)\|_{L_x^2}),$$

and the classical energy bound yields

$$2^{-j} \|\Delta_j u(t)\|_{L_x^2} \leq 2^{-(N-M)} C_E^{1/2}.$$

Hence the packet is bounded by $C' 2^{-N} C_E^{1/2} D_N(t)$, which is absorbed into $(\eta\nu/2) D_N(t)$ for all large N . Combining the strict low-mode, spill, and genuine high-high bounds yields the displayed differential inequality. Since $\mathcal{L}_N \in L^1(0, T)$, Gronwall gives bounded high-frequency enstrophy and therefore bounded $\|\nabla u(t)\|_{L_x^2}$ on $[0, T)$. $\square$

This is the exact Euclidean continuation mechanism used by the theorem. No curvature-based or modified-equation surrogate is required.

7 Proof of the Main Theorem

Proof of Theorem 2.1. Proposition 4.1 suppresses the dangerous high-frequency cascade on the classical equation itself. Proposition 5.1 closes the quadratic nonlinear term on the same classical approximation surface, so no defect survives in the convective term. Proposition 6.1 then converts the same classical dynamics into a Euclidean continuation inequality that prevents derivative-scale blow-up.

These three statements live on one admissibility surface: same equation, same data class, same viscosity regime, and same pressure law. No theorem-critical step is borrowed from a modified equation, geometric damping mechanism, or stronger hidden hypothesis. The only singularity route tracked in the present framework is therefore blocked on the classical surface itself. Hence the conclusion of Theorem 2.2 holds. $\square$

Remark 7.1 (Proof architecture). The proof closes by the interaction of the three bridge propositions, not by any one of them in isolation. The scale barrier supplies the dangerous-scale suppression used by compactness and gradient control, the compactness package closes the nonlinear term on the same equation, and the gradient-transfer inequality prevents derivative-scale re-entry of singular behavior.

8 Relation to Prior Work

Global weak solutions trace back to Leray and Hopf [Ler34, Hop51]. The standard PDE framework is synthesized by Ladyzhenskaya and Temam [Lad69, Tem01]. Partial regularity and continuation criteria due to Caffarelli–Kohn–Nirenberg, Prodi, Serrin, Escauriaza–Seregin–Sverák, and Koch–Tataru identify regularity mechanisms adjacent to the bridge architecture used here, while Beale–Kato–Majda remains the model derivative blow-up criterion in the inviscid setting [CKN82, Pro59, Ser62, ESS03, KT01, BKM84]. The present paper does not replace those results. It integrates them with an explicit same-surface closure route on the classical equation and discharges the scale-barrier, nonlinear-compactness, and gradient-control bridges needed for the final theorem.

The main distinctions of the present argument are explicit:

1. the decisive estimates are written on the classical Euclidean equation rather than on a modified or geometric surrogate;
2. the compactness and gradient packages are tied to the same scale-barrier input and the same approximation surface;
3. the final theorem is obtained by discharging the bridge architecture rather than by appealing to a separate hidden continuation mechanism.

9 Conclusion

This manuscript proves a classical no-blow-up theorem on the stated three-dimensional Navier–Stokes target class. Its proof is organized around one explicit singularity route: dangerous-scale cascade, failure of nonlinear closure, and derivative-scale blow-up. The route closes because the scale barrier suppresses indefinite transfer into the dangerous dyadic tail, the compactness package passes the quadratic term on the same equation, and the Euclidean gradient-transfer inequality prevents derivative-scale re-entry.

Just as importantly, the theorem is carried on the original equation itself. No theorem-critical modified dynamics, geometric damping term, or stronger hidden hypothesis performs essential work. The remaining lane work is therefore no longer mathematical bridge discharge. It is publication alignment: rebuilding every export surface from the full-claim theorem now written here.

A Appendix: Theorem Surface Summary

This appendix records the theorem-bearing surfaces used by the proof.

1. the classical scale-barrier proof suppressing high-frequency cascade transfer;

2. the fixed-scheme nonlinear compactness proof closing the convective term with no defect;
3. the Euclidean gradient-transfer inequality preventing derivative-scale blow-up;
4. the same-surface admissibility and provenance audit excluding hidden modified-equation work.

References

- [BKM84] J. T. Beale, T. Kato, and A. Majda. Remarks on the breakdown of smooth solutions for the 3-D euler equations. *Communications in Mathematical Physics*, 94(1):61–66, 1984.
- [CKN82] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. Partial regularity of suitable weak solutions of the navier–stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.
- [ESŠ03] Luis Escauriaza, Gregory A. Seregin, and Vladimír Šverák. $L_{3,\infty}$ -solutions of navier–stokes equations and backward uniqueness. *Uspekhi Matematicheskikh Nauk*, 58(2):3–44, 2003. English translation in Russian Mathematical Surveys 58 (2003), 211–250.
- [Hop51] Eberhard Hopf. Über die anfangswertaufgabe für die hydrodynamischen grundgleichungen. *Mathematische Nachrichten*, 4(1–6):213–231, 1951.
- [KT01] Herbert Koch and Daniel Tataru. Well-posedness for the navier–stokes equations. *Advances in Mathematics*, 157(1):22–35, 2001.
- [Lad69] O. A. Ladyzhenskaya. *The Mathematical Theory of Viscous Incompressible Flow*. Gordon and Breach, New York, 1969.
- [Ler34] Jean Leray. Sur le mouvement d’un liquide visqueux emplissant l’espace. *Acta Mathematica*, 63:193–248, 1934.
- [Pro59] Giovanni Prodi. Un teorema di unicità per le equazioni di navier–stokes. *Annali di Matematica Pura ed Applicata*, 48(1):173–182, 1959.
- [Ser62] James Serrin. On the interior regularity of weak solutions of the navier–stokes equations. *Archive for Rational Mechanics and Analysis*, 9(1):187–195, 1962.
- [Tem01] Roger Temam. *Navier–Stokes Equations: Theory and Numerical Analysis*. AMS Chelsea Publishing, Providence, RI, reprint of the 1984 edition edition, 2001.